



Acute Stabilization Hypoglycemia: Outpatient Pathway

Inclusion

Greater than 1 year of age (can consider for age < 1 year old with age-appropriate PO options, consider taking history if stable and consulting pediatric hospitalist/pediatric endocrinology for this age group)

Note:

In a child with a likely known etiology for hypoglycemia, or diagnosis predisposing to hypoglycemia, consider discussing with the appropriate subspecialist (ie: Endocrinology, Metabolic/Genetics, etc. for diagnosis-specific additional treatment options.)

*****If unknown etiology, try to obtain critical sample labs first prior to treatment**

SEE LAB ORDERS ON PAGE 2

Tubes required

- Gold
- Gray
- Red

Clinical suspicion of hypoglycemia
(sweating, tremors, palpitations, tachycardia, hunger)

Check POCT glucose

GLUCOSE < 50mg/dl

Able to tolerate PO

P.O. options: (give half this dose if child is < 2 yo)

- 1) 4 ounces fruit juice
- 2) 4 ounces of regular soda
- 3) 2.5 tablespoons sugar
- 4) 4 glucose tablets (4 gm carb each)
- 5) 1 small tub cake mate frosting
- 6) 2 rolls smarties candy
- 7) 40% dextrose gel**

Recheck blood sugar in 10-15 minutes

2nd glucose < 50mg/dl

Maintain blood glucose > 70 and give complex carbs (crackers, peanut butter, milk). Can consider dispo home if tolerating PO well and maintaining sugar > 70

Unable to tolerate PO well

Insert IV
(if available)

Give **D10W**
bolus (2 ml/kg)

Continue
D5NS at 1xM

Admit for
observation

**Consult
specialists and
Transfer to ED**

UNSTABLE (lethargy, confusion, irritability, decreased consciousness, seizure)

Ensure adequate AIRWAY, BREATHING, CIRCULATION. Connect to monitors if available.

Check POCT glucose if glucose < 50

Alert and airway intact

Attempt PO options

++Consider glucagon for indications listed on page 2

****Consider 40% dextrose gel if available (dosing on page 2)**

Not Alert

Glucagon if available

IM:
0.5 mg (< 20 kg)
or 1 mg (> 20 kg)

Nasal (≥ 4 yo):
3mg in single nostril

****consider dextrose gel if available**

Call 911 and activate EMS

***** LAB ORDERS****1ST PRIORITY**

- Serum glucose (GRAY)
- Serum beta hydroxybutyric acid (GOLD)
- Serum insulin (GOLD)
- Serum cortisol (GOLD)
- Serum lactate (GRAY)
- Serum electrolytes (GOLD)
- Serum growth hormone (RED)
- Serum FFA (RED)

2ND PRIORITY

- Serum ammonia (LIGHT GREEN)
- plasma acylcarnitine profile (GREEN)
- Serum total and free carnitine (GREEN)
- Serum C-peptide (GOLD)

3RD PRIORITY (not required at this time)

- Urine organic acids
- UA (for urine ketones)
- Plasma amino acids
- TSH/free T4

++Glucagon indications

- Known diabetic patients on insulin who cannot tolerate PO or seizing/unresponsive
- Sulfonylurea ingestion
- Hyperinsulinism
- Insulin overdose

**** Dextrose gel dosing**

- Single dose 0.2 gm/kg for infants and young children
- Max dose 15 gm



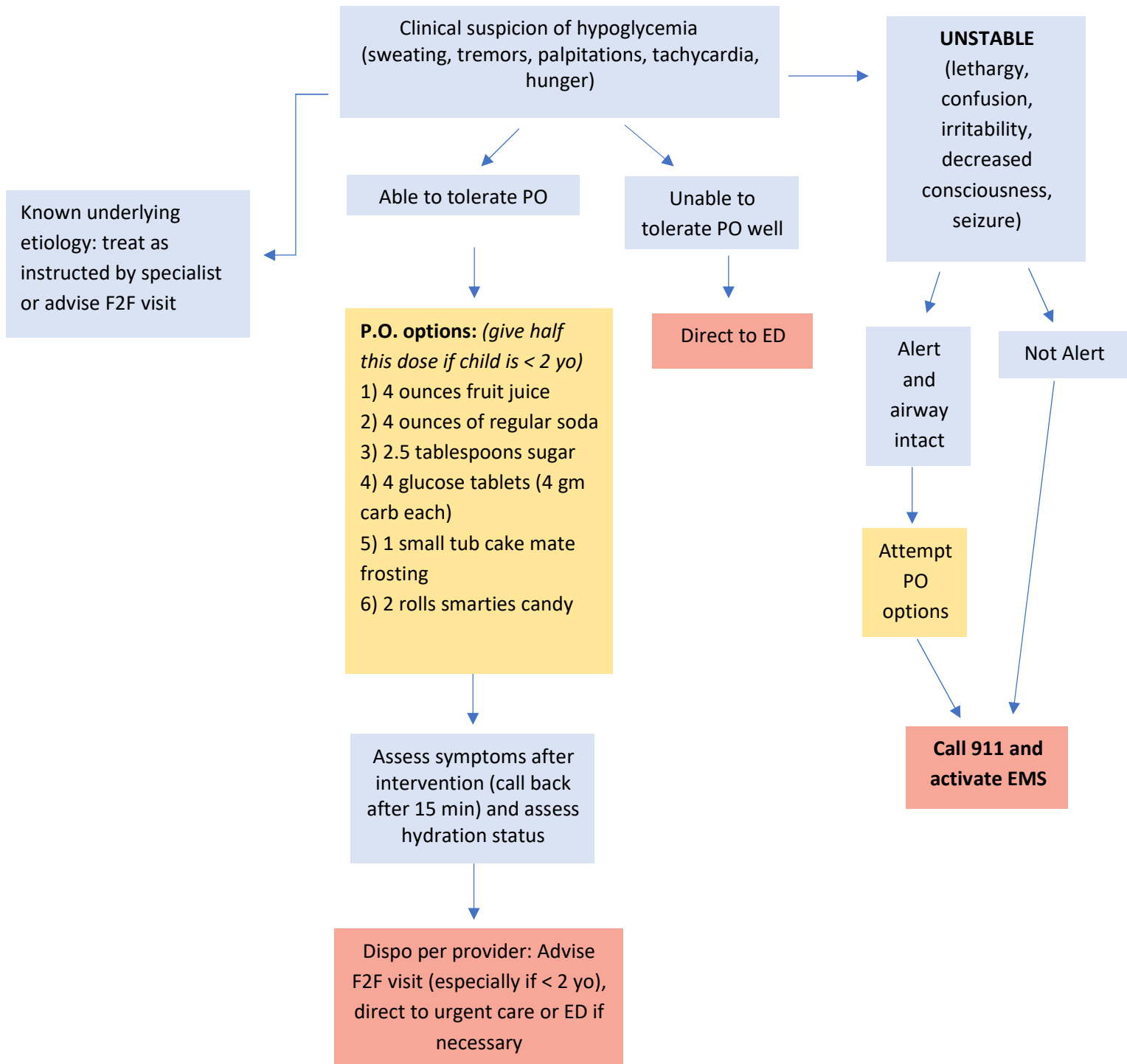
Acute Stabilization Hypoglycemia: VMC Pathway

Inclusion

Greater than 1 year of age with clinical symptoms of gastroenteritis or suspicion of hypoglycemia symptoms (*can consider for age < 1 year old with age-appropriate PO options, but highly recommend advising F2F visit and take appropriate history if stable*)

Note:

In a child with a likely known etiology for hypoglycemia, or diagnosis predisposing to hypoglycemia, consider discussing with the appropriate subspecialist (ie: Endocrinology, Metabolic/Genetics, etc. for diagnosis-specific additional treatment options.)





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References:

Abraham MB, Jones TW, Naranjo D, et al. ISPAD Clinical Practice Consensus Guidelines 2018: Assessment and management of hypoglycemia in children and adolescents with diabetes. *Pediatr Diabetes*. 2018;19 Suppl 27:178-192. doi:10.1111/pedi.12698

De-Leon Crutchlow, D. D., MD, & Lord, K., MD. (2019). Diagnostic approach to hypoglycemia in infants and children. *Uptodate*. Retrieved July 27, 2020, from <https://www.uptodate.com/contents/diagnostic-approach-to-hypoglycemia-in-infants-and-children?search=hypoglycemia>

Thornton PS, Stanley CA, De Leon DD, et al. Recommendations from the Pediatric Endocrine Society for Evaluation and Management of Persistent Hypoglycemia in Neonates, Infants, and Children. *J Pediatr* 2015; 167:238.